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RESEARCH PAPER

Point-based Spatio-temporal Convolution and Dual-channel Attention for Lightweight Multimodal Deepfake Detection

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Abstract The rapid advancement of generative models has led to the widespread proliferation of highly realistic deepfake videos, posing serious threats to digital media integrity, public trust, and online security. Although recent deepfake detection methods have achieved high detection accuracy, most rely on computationally intensive architectures that limit their applicability in real-time and resource-constrained deployment scenarios. This creates a critical gap between academic performance and practical usability. This study proposes an efficient multimodal deepfake detection framework that jointly models spatial-temporal visual artifacts and audio-visual correlation inconsistencies while explicitly considering computational efficiency. The proposed method integrates a Fast Point Spatio-Temporal Convolution module for redundancy-aware visual feature extraction, a Dual-Channel Attention mechanism to separately capture intra-modal artifacts and cross-modal inconsistencies, and an attention-guided cross-modal fusion strategy for adaptive multimodal integration. An Efficiency-Aware Multi-Objective Bayesian Optimization framework is employed to tune key hyperparameters by jointly optimizing detection performance, computational complexity, and inference latency. Extensive experiments on FaceForensics++, DFDC, and Celeb-DF datasets demonstrate that the proposed framework achieves state-of-the-art detection performance, attaining up to 98.5% accuracy on FaceForensics++ while maintaining low computational overhead with 14.2 GFLOPs and 28 ms inference latency per video clip. The framework outperforms existing CNN- and transformer-based methods in terms of the accuracy-efficiency trade-off and exhibits strong generalization across datasets. This study demonstrates that robust multimodal deepfake detection can be achieved without sacrificing computational efficiency. By integrating efficient feature extraction, principled multimodal attention, and deployment-aware hyperparameter optimization into a unified framework, the proposed method provides a practical and scalable solution suitable for real-world deepfake detection applications.

Keywords Deepfake detection · Multimodal learning · Audio-visual fusion · Attention mechanisms · Computational efficiency · Bayesian optimization · Real-time deployment

1 Introduction

The rapid advancement of deep generative models has enabled the creation of highly realistic synthetic media, commonly referred to as deepfakes (George and George 2023). These manipulated videos, often indistinguishable from authentic content to the human eye, pose significant threats to digital media integrity, public trust, and societal stability (Sheikh et al. 2024). Deepfakes have been increasingly exploited in malicious contexts, including misinformation campaigns, identity impersonation, financial fraud, and non-consensual content creation (Mansoor 2024). As the accessibility of deepfake generation tools continues to grow, the urgency for reliable and scalable detection mechanisms has become a critical research and societal concern (Khimi et al. 2024). Recent years have witnessed substantial progress in deepfake detection methods, particularly through the adoption of deep learning techniques (Khan et al. 2025). Many state-of-the-art approaches focus on maximizing detection accuracy by employing deep convolutional neural networks, transformer-based architectures,

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or complex multimodal fusion strategies (Abdulmaksoud and Ahmed 2025). While these methods demonstrate impressive performance on benchmark datasets, they often rely on computationally intensive architectures that demand substantial memory, processing power, and inference time (Menghani 2023). As a result, such models are difficult to deploy in real-world environments where latency, scalability, and hardware constraints are critical considerations (Tyagi 2025). Computational cost has therefore emerged as a major barrier to the practical adoption of deepfake detection systems (Kaludewage 2025). Applications such as online content moderation, live video verification, and edge-based media analysis require detectors that can operate in real time or near real time, often on resource-constrained platforms (Shao et al. 2025). Models optimized solely for accuracy without considering efficiency are ill-suited for these deployment scenarios, highlighting the need for detection frameworks that balance effectiveness with computational feasibility (Villafranca et al. 2025).

Despite the availability of highly accurate deepfake detection models, a fundamental challenge remains in balancing detection performance with computational efficiency (Wang et al. 2024a). The accuracy-efficiency trade-off refers to the inherent tension between achieving high detection accuracy and maintaining low computational cost, including reduced FLOPs, inference latency, and memory usage (Joshua 2025). Many existing approaches implicitly assume access to powerful hardware and unconstrained inference time, leading to models that perform well in controlled experimental settings but fail to scale in real-world deployments. Multimodal deepfake detection methods, which jointly analyze visual and audio information, further exacerbate this challenge (Liu et al. 2024a). While multimodal fusion improves robustness by leveraging complementary cues, it often introduces additional computational overhead through complex attention mechanisms, deep encoders, or transformer-based fusion modules. Consequently, existing multimodal models are typically unsuitable for real-time or edge deployment, where strict latency and energy constraints must be met. This gap between academic performance and practical usability motivates the need for a detection framework that explicitly incorporates efficiency considerations into both architectural design and optimization.

The primary objective of this study is to develop a multimodal deepfake detection framework that operates effectively under strict computational constraints while maintaining competitive detection accuracy. Specifically, this work aims to design an efficient architecture that minimizes redundant computation in spatio-temporal visual processing and avoids unnecessary complexity in multimodal fusion. In parallel, the study seeks to preserve strong detection performance across diverse manipulation techniques, compression levels, and real-world recording conditions. Another key objective is to enable real-time and resource-constrained deployment. The proposed framework is intended to be suitable for practical applications such as online content moderation, media authentication, and edge-based video analysis. To achieve this, computational efficiency is treated as a first-class design criterion rather than a secondary consideration, ensuring that the resulting model is both effective and deployable.

This study makes the following key contributions to the field of deepfake detection:

1. An efficient fast point spatio-temporal convolution module that reduces redundant computation by operating on representative spatial points while preserving discriminative visual features associated with facial manipulation artifacts.
2. A dual-channel attention mechanism that explicitly separates intra-modal spatial-temporal artifact modeling from inter-modal audio-visual inconsistency analysis, enabling robust detection across diverse manipulation scenarios.
3. A comprehensive accuracy-efficiency evaluation across multiple benchmark datasets, demonstrating that the proposed framework achieves state-of-the-art detection performance while maintaining low computational overhead suitable for real-time deployment.

The remainder of this paper is organised as follows. Section 2 reviews related work on deepfake detection, with a focus on spatio-temporal modeling, multimodal fusion, attention mechanisms, and efficiency-aware approaches. Section 3 presents the proposed methodology in detail, including the system architecture, fast point spatio-temporal convolution, dual-channel attention mechanism, cross-modal fusion strategy, and efficiency-aware hyperparameter optimization. Section 4 reports the experimental results and provides a detailed analysis of detection performance and computational efficiency, and discusses the implications of the results, design trade-offs, and practical deployment considerations. Finally, Sect. 5 concludes the paper and outlines directions for future research.

2 Literature Review

Rossler et al. (2019) introduced one of the earliest large-scale benchmarks for face manipulation detection, providing manipulated videos generated using multiple techniques and compression levels. The study systematically evaluated CNN-based detectors and demonstrated that compression significantly degrades detection performance, highlighting

the importance of robustness. While the dataset standardized evaluation protocols and accelerated progress in the field, the work primarily focused on benchmarking rather than proposing a computationally efficient or multimodal detection framework. Dolhansky et al. (2020) presented the DeepFake Detection Challenge dataset, consisting of over 100,000 videos with diverse subjects, environments, and manipulation artifacts. The study showed that models trained on large and diverse datasets exhibit improved robustness to real-world conditions. However, the work emphasized dataset creation and evaluation rather than addressing architectural efficiency or real-time deployment constraints, leaving optimization under resource limitations largely unexplored. Li et al. (2020) proposed a high-quality deepfake dataset designed to challenge detectors that perform well on earlier benchmarks. Their analysis revealed that many existing models suffer significant performance degradation when evaluated on Celeb-DF, underscoring generalization limitations. While the dataset advances realism and evaluation rigor, it does not address how detection models can be designed to remain efficient and scalable in practice. Wang et al. (2022) introduced a multi-scale transformer-based architecture for deepfake detection that captures subtle artifacts across spatial resolutions. The model achieved strong detection performance by leveraging transformer attention mechanisms and multimodal cues. Despite its accuracy, the transformer-heavy design incurs substantial computational overhead, making real-time or edge deployment challenging and motivating the need for more efficiency-aware alternatives.

Oorloff et al. (2024) proposed an audio-visual feature fusion framework that leverages complementary masking and cross-modal learning to improve robustness against unseen manipulations. The study demonstrated that explicit audio-visual modeling improves generalization across datasets. However, the two-stage learning process and multimodal fusion introduce additional training and inference complexity, which may limit deployment in constrained environments. Nguyen et al. (2024) introduced a localized artifact attention network that focuses on artifact-prone facial regions using hierarchical attention mechanisms. The model achieved quality-agnostic and generalizable detection across compression levels. While LAA-Net improves robustness, its reliance on multiple attention stages increases inference cost, and efficiency optimization is not explicitly addressed. Nie et al. (2024) proposed a forgery-aware audio-distilled learning framework that injects audio-informed forgery cues into visual encoders. The approach improved cross-dataset performance by aligning audio and visual forgery signals. Nevertheless, the method depends on careful distillation strategies and pretrained backbones, adding architectural and training complexity. Yang et al. (2023) presented a joint audio-visual learning framework that explicitly models inconsistencies between speech and facial motion. The study showed that multimodal inconsistency cues significantly improve detection accuracy, particularly for lip-sync manipulations. However, the approach is sensitive to noisy or missing audio and does not explicitly consider computational efficiency.

Zhang et al. (2024) introduced a contrastive learning-based framework for audio-visual alignment, followed by attention-driven multimodal detection. Their results demonstrated improved robustness and generalization by learning aligned multimodal representations. The reliance on large-scale contrastive pretraining, however, increases computational demands and training complexity. Liu et al. (2024b) focused on detecting lip-sync forgeries by modeling temporal inconsistencies between lip movements and audio signals. The method achieved strong performance for speech-driven deepfakes by exploiting fine-grained temporal cues. Its specialization to lip-region artifacts limits its applicability to more general face manipulation scenarios. Qi et al. (2017) proposed a hierarchical point-based neural architecture for learning from unordered point sets, demonstrating that representative sampling and local aggregation can significantly reduce redundancy while preserving discriminative power. Although originally designed for 3D point clouds, the underlying principles motivate point-based processing strategies for reducing spatial redundancy in video-based deepfake detection. Karl et al. (2023) reviewed multi-objective optimization techniques for balancing accuracy and resource constraints in machine learning models. The study highlighted Bayesian optimization as an effective and principled approach for navigating trade-offs between performance and efficiency. While comprehensive, the survey does not address application-specific integration into deepfake detection pipelines.

Miao et al. (2021) proposed a local-feature-based transformer framework that emphasizes patch-level relationships to improve generalization in face manipulation detection. Their results showed that focusing on local artifacts enhances robustness across datasets. However, the approach does not explicitly model temporal dynamics and remains computationally intensive. Wang and Huang (2024b) introduced an articulatory representation-based audio-visual detection framework that captures speech-face dynamics using linguistically inspired features. The method demonstrated improved robustness for audiovisual deepfakes. Its reliance on high-quality audio and precise alignment limits effectiveness under noisy or low-resource conditions. Liang et al. (2024) proposed self-supervised audio-visual speech representation learning to enhance forgery detection robustness. By learning speech-aware multimodal embeddings, the method achieved strong cross-dataset generalization.

However, the approach requires large-scale pretraining and significant computational resources, posing challenges for lightweight deployment.

Chakrabarti and De (2025) proposed a three-stream feature-level fusion framework for multimodal emotion recognition that jointly processes vocal and facial expression cues, demonstrating significant accuracy gains through explicit cross-modal integration. While the study establishes the value of multimodal fusion in affective computing, it does not address the computational efficiency of the fusion pipeline, which is a central concern in our work. Anitha and Selvy (2026) introduced a multi-context squeeze-excitation framework with explainable attention for cervical spine fracture detection in CT imaging, demonstrating that channel-wise attention mechanisms can enhance feature discrimination in medical image analysis tasks while maintaining interpretability. However, attention is applied within a single visual modality and deployment efficiency is not explicitly optimized. Singh and Sharmin (2026) proposed HybridDerainNet, a lightweight dual-stream CNN-transformer network for single image deraining, demonstrating that efficient hybrid architectures can deliver competitive performance with reduced computational overhead. While focused on image restoration rather than detection, the architectural principles of dual-stream lightweight design are directly relevant to the efficiency-aware objectives of our framework.

Recent progress in deepfake detection has been largely driven by deep learning-based visual and multimodal approaches aimed at identifying subtle manipulation artifacts in face-centric videos. Benchmark datasets such as FaceForensics++, DFDC, and Celeb-DF have standardized evaluation protocols and revealed critical challenges related to compression, real-world variability, and generalization. To improve detection accuracy, prior research has explored deep convolutional architectures, transformer-based models, and multimodal fusion strategies that leverage audio-visual cues such as lip-speech synchronization and cross-modal inconsistencies, often augmented with attention mechanisms and localized artifact modeling. While these methods achieve strong performance on benchmark datasets, they typically prioritize accuracy without explicitly addressing computational efficiency. As a result, transformer-heavy designs, multi-stage fusion pipelines, and contrastive pretraining strategies introduce substantial computational overhead, limiting real-time and resource-constrained deployment. Furthermore, hyperparameter tuning in existing studies is predominantly accuracy-driven, relying on manual or heuristic search methods that ignore inference latency and computational cost. Although multi-objective optimization has been studied in broader machine learning contexts, its application to multimodal deepfake detection remains limited, highlighting a persistent gap between state-of-the-art detection performance and practical deployability.

2.1 Research Gap

Based on the literature review, several critical research gaps can be identified. First, there is a lack of deepfake detection frameworks that explicitly balance detection accuracy with computational efficiency, particularly in multimodal settings where fusion mechanisms often increase inference cost. Existing approaches tend to prioritize performance on benchmark datasets without ensuring feasibility for real-time or edge deployment. Second, while multimodal deepfake detection has shown clear advantages over unimodal methods, current architectures frequently rely on complex attention or transformer-based designs that are computationally expensive and difficult to scale. Efficient exploitation of spatial redundancy in face-centric videos and targeted modeling of artifact-prone regions remain underexplored. Third, the integration of efficiency-aware optimization into the model development process is largely absent in existing studies. Hyperparameter tuning is commonly treated as an auxiliary step focused solely on accuracy, rather than as a multi-objective problem that jointly considers performance, computational complexity, and inference latency. Finally, there is limited work on unified frameworks that combine efficient spatio-temporal visual processing, principled multimodal attention, adaptive fusion, and deployment-aware optimization within a single end-to-end system. This gap highlights the need for a holistic approach that bridges the divide between academic performance and real-world applicability.

The proposed FPDCA-CNN framework differs from existing methods along three key technical dimensions. First, unlike point-based architectures such as PointNet++ (Qi et al. 2017), which process 3D point clouds for object recognition, the proposed Fast Point Spatio-Temporal Convolution module adapts point-based processing to 2D facial video frames, specifically targeting spatial redundancy in face-centric deepfake detection. The sampling strategy is guided by artifact saliency rather than geometric regularity, and temporal dynamics are modeled through factorized 1D temporal convolution rather than hierarchical set abstraction. Second, unlike dual-attention designs in related domains such as Anitha and Selvy (2026) and Chakrabarti and De (2025), which apply attention within a single modality or fuse features post-hoc, the proposed Dual-Channel Attention mechanism explicitly separates intra-modal spatial-temporal artifact detection from inter-modal audio-visual correlation modeling, with learnable adaptive weighting between the two pathways. This decoupled

design directly addresses the heterogeneous artifact profiles observed across FaceForensics++, DFDC, and Celeb-DF. Third, while multi-objective optimization has been studied in broader machine learning contexts (Karl et al. 2023) and lightweight architectures have been explored for efficiency in related tasks (Singh and Sharmin 2026), the proposed Efficiency-Aware Multi-Objective Bayesian Optimization framework is the first to integrate deployment constraints, specifically FLOPs and inference latency budgets, directly into hyperparameter tuning for multimodal deepfake detection. These design choices collectively distinguish the proposed framework from existing work and provide a principled, deployment-ready solution.

3 Proposed Methodology

3.1 System Overview

The proposed framework is an end-to-end multimodal deepfake detection system that jointly analyzes visual manipulation artifacts and audio-visual inconsistencies under computational constraints. Given an input video, synchronized visual frames and audio signals are extracted, and facial regions are detected and aligned to normalize variations in pose, scale, and recording conditions. Each video is segmented into short temporal clips using uniform sampling to capture dynamic facial behavior while maintaining computational efficiency, and corresponding audio segments are temporally aligned with the visual clips.

Visual features are extracted using a fast point spatio-temporal convolution module that operates on representative spatial points within facial regions, reducing redundant computation while preserving discriminative spatial-temporal cues. Audio features are obtained using a lightweight convolutional network that captures temporal speech characteristics. The extracted features are refined through a dual-channel attention mechanism, where one channel emphasizes intra-modal spatial-temporal visual artifacts and the other models inter-modal audio-visual correlations to identify synchronization inconsistencies Fig. 1.

The refined audio and visual representations are fused through an attention-guided cross-modal fusion strategy and passed to a lightweight classification head to produce clip-level predictions, which are aggregated for video-level decision making. To ensure suitability for real-time and resource-constrained deployment, key architectural and training hyperparameters are optimized using an efficiency-aware multi-objective Bayesian optimization framework that jointly considers detection performance and computational cost. This design enables robust multimodal deepfake detection across diverse datasets while maintaining practical inference efficiency.

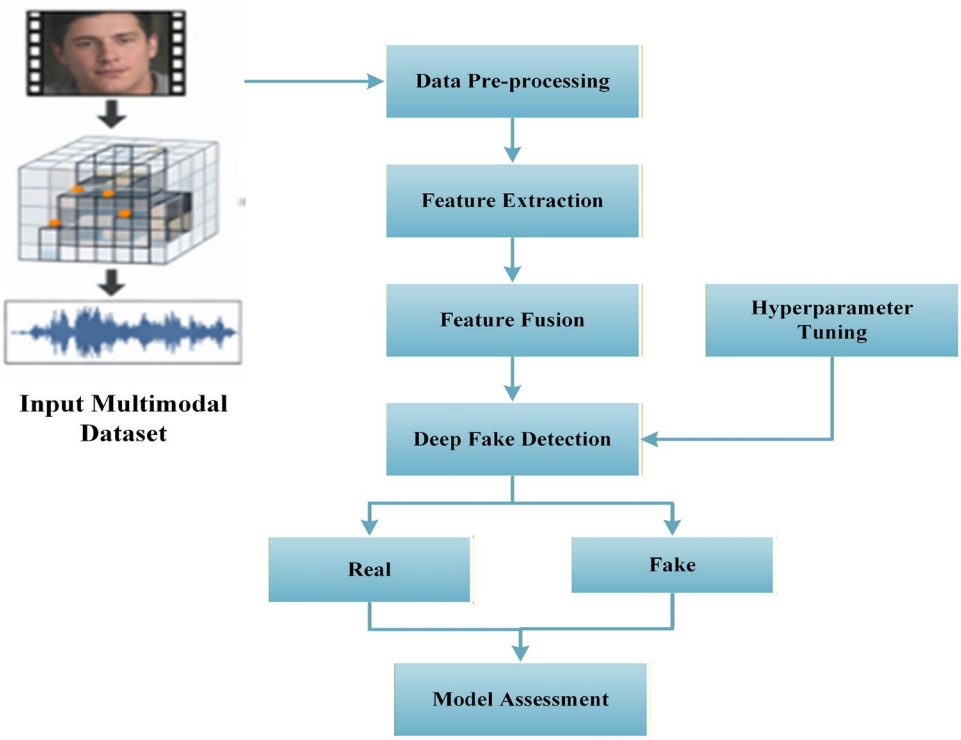


Fig. 1 Workflow of the proposed framework

3.2 Data Collection and Evaluation Protocols

To ensure a comprehensive and unbiased evaluation, this study employs three publicly available benchmark datasets that are widely adopted in deepfake detection research: FaceForensics++ (ondyari 2026), DFDC (Meta AI 2026), and Celeb-DF (Li 2026). These datasets were selected to cover a broad range of manipulation techniques, compression levels, and real-world recording conditions, thereby supporting both in-dataset performance evaluation and cross-dataset generalization analysis. FaceForensics++ is used as the primary dataset for model training and validation. It contains 1,000 pristine videos and 4,000 manipulated videos generated using four common face manipulation techniques: Deepfakes, Face2Face, FaceSwap, and NeuralTextures. Each video is provided at three compression levels—raw (c0), medium compression (c23), and high compression (c40)—to simulate varying real-world video quality conditions. Following the official evaluation protocol, the dataset is split into 720 training videos, 140 validation videos, and 140 test videos, ensuring fair and reproducible comparison with existing methods.

DFDC (DeepFake Detection Challenge) is employed to evaluate large-scale robustness under diverse and realistic conditions. The dataset consists of over 119,000 videos collected from 3,426 participants, featuring substantial variability in lighting, backgrounds, camera devices, and compression artifacts. Due to its scale and diversity, DFDC is used primarily for training and large-scale performance validation, reflecting real-world deployment scenarios.

Celeb-DF is utilized exclusively for cross-dataset generalization assessment. It includes 5,639 videos, comprising 590 real and 5,049 fake samples, with high-quality face synthesis targeting celebrity identities. Since Celeb-DF exhibits fewer obvious artifacts and higher visual realism than FaceForensics++, it provides a challenging benchmark for evaluating the generalization capability of models trained on other datasets.

Across all datasets, evaluation protocols adhere to established practices in recent spatio-temporal deepfake detection studies. Temporal clips are sampled consistently across videos, and identical preprocessing and evaluation settings are applied to all comparative methods to ensure fairness.

Table 1 Dataset statistics summary

Dataset	Total videos	Real	Fake	Primary use
FaceForensics++	5,000	1,000	4,000	Training/ Validation
DFDC	119,154	23,654	95,500	Training/Large-scale Eval
Celeb-DF	5,639	590	5,049	Cross-dataset Evaluation
Total	129,793	25,244	104,549	—

Performance is reported using standard metrics, including AUC and inference latency, enabling both accuracy-based and efficiency-based comparisons. Table 1 summarises the key statistics of the datasets used in this study.

3.2.1 Data Pre-processing

Let a video sample be denoted as $\mathcal{V} = \{I_t, A_t\}_{t=1}^{T}$, where $I_t \in \mathbb{R}^{H\times W\times 3}$ represents the RGB frame at time t, A_t denotes the corresponding raw audio signal, and T is the total number of frames. Data pre-processing aims to transform raw audio-visual inputs into temporally aligned and normalized representations suitable for efficient multimodal learning across datasets with heterogeneous characteristics. For the visual modality, a face detection function $\mathcal{F}(\cdot)$ is applied to each frame to localize the facial region of interest. The detected face is then geometrically aligned using facial landmarks and resized to a fixed spatial resolution $H' \times W'$. This process can be expressed as

$$\hat{I}_t = \mathcal{R}(\mathcal{A}(\mathcal{F}(I_t))),$$

where $\mathcal{A}(\cdot)$ denotes facial alignment and $\mathcal{R}(\cdot)$ represents spatial resizing. This normalization reduces variations caused by pose, scale, and camera motion, which are particularly prominent in DFDC and compressed FaceForensics++ videos. To capture temporal information while maintaining computational efficiency, each video is segmented into short clips of length L frames using uniform temporal sampling. Let τ_k denote the sampling index for the k-th clip; the resulting visual clip is defined as

$$X_k = \{\hat{I}_{\tau_k+i}\}_{i=0}^{L-1}.$$

This strategy ensures consistent temporal coverage across videos of varying durations and allows effective modeling of motion artifacts and temporal inconsistencies introduced by deepfake generation processes. For the audio modality, the raw audio signal A_t is extracted from the video and resampled to a fixed sampling rate. The audio stream is segmented into temporal windows aligned with the visual clips to preserve synchronization. Given an audio segment a_k corresponding to clip X_k, a time-frequency representation is computed using a short-time Fourier transform (STFT),

$$S_k(f,\tau) = \left| \sum_n a_k(n)\, w(n-\tau)\, e^{-j2\pi f n} \right|,$$

where $w(\cdot)$ is a windowing function. The resulting spectrogram is normalized to reduce the effects of amplitude variation and background noise, yielding

$$\hat{S}_k = \frac{S_k - \mu_S}{\sigma_S},$$

with μ_S and σ_S denoting the mean and standard deviation computed over the training set. To ensure precise multimodal correspondence, visual clips and audio segments are temporally synchronized such that each training sample is represented as a paired input $(X_k, \hat{S}_k)$. Each pair inherits the ground-truth label $y \in \{0, 1\}$ from the source video, where 0 and 1 indicate real and fake content, respectively. During training, clip-level predictions are learned independently, while video-level decisions are obtained through aggregation of clip-level outputs.

This unified pre-processing pipeline standardizes spatial, temporal, and acoustic representations across datasets with varying compression levels, recording conditions, and manipulation quality. By enforcing consistent alignment and synchronization, the pre-processing stage facilitates effective multimodal feature learning while supporting the computational efficiency objectives of the proposed framework.

3.3 Fast Point Spatio-Temporal Convolution Module

The Fast Point Spatio-Temporal Convolution Module is designed to efficiently extract discriminative visual features from face-centric videos while mitigating the high computational cost associated with dense spatio-temporal convolutions. In widely used deepfake datasets such as FaceForensics++, DFDC, and Celeb-DF, video frames exhibit substantial spatial redundancy, particularly within facial regions where neighbouring pixels often share similar appearance under both authentic and manipulated conditions. Consequently, applying standard 3D convolutions uniformly over entire frame volumes leads to significant redundant computation with limited improvement in detection performance. To address this inefficiency, the proposed module adopts a point-based spatio-temporal processing strategy that focuses computation on a subset of representative spatial locations while preserving essential temporal dynamics. Let a temporally sampled video clip be denoted as

$$X = \left\{ I_t \in \mathbb{R}^{H \times W \times C} \right\}_{t=1}^{T},$$

where H and W denote the spatial resolution, C is the number of channels, and T is the number of frames. Instead of processing all spatial locations, the proposed module samples a set of N representative spatial points from each aligned facial frame. The sampled point set at time t is defined as

$$\mathcal{P}_t = \left\{ p_t^i \in \mathbb{R}^2 \right\}_{i=1}^{N},$$

where each point corresponds to a spatial location within the facial region. A distance-aware sampling strategy is employed to ensure spatial diversity and adequate coverage of artifact-prone areas such as facial boundaries, eyes, and mouth regions, which are particularly susceptible to synthesis errors across all three datasets. This targeted sampling significantly reduces spatial redundancy while retaining informative local regions. It is important to note that spatial points are sampled independently per frame using the distance-aware strategy, and no explicit cross-frame point correspondence is enforced at the sampling stage. Temporal consistency is instead achieved implicitly at the feature aggregation stage: each sampled point index generates a feature vector f_t^i at every frame t, and the temporal sequence $F^i = \{f_t^i\}_{t=1}^T$ for that index is processed by a one-dimensional temporal convolution $\phi_t(\cdot)$. This factorised design captures motion patterns and temporal inconsistencies across frames without requiring explicit point tracking, which would introduce additional computational cost. Furthermore, since each point aggregates features within a local neighbourhood $\mathcal{N}(p_t^i)$ of radius r rather than operating on a single pixel, the risk of significant spatial context loss is mitigated: the neighbourhood structure preserves local spatial relationships and texture patterns even when individual point locations vary slightly across frames. For each sampled point p_t^i, a local spatial neighbourhood $\mathcal{N}(p_t^i)$ is constructed within a fixed radius r. Local point-wise features are then computed as

$$f_t^i = \max_{q \in \mathcal{N}(p_t^i)} \phi_s\left(I_t(q) - I_t(p_t^i)\right),$$

where $\phi_s(\cdot)$ denotes a lightweight spatial convolution followed by normalization and non-linearity. Max pooling is used to ensure permutation invariance within each local neighbourhood. This operation effectively captures fine-grained spatial distortions such as blending artifacts, texture inconsistencies, and boundary irregularities commonly observed in manipulated frames. To model temporal dynamics, spatial features corresponding to the same sampled point across consecutive frames are aggregated along the temporal dimension. Let

$$F^i = \left\{ f_t^i \right\}_{t=1}^{T}$$

denote the temporal feature sequence for point . Temporal convolution is applied as

$$\hat{F}^i = \phi_t(F^i),$$

where $\phi_t(\cdot)$ is a one-dimensional temporal convolution that captures motion patterns and temporal inconsistencies such as jittering, unnatural transitions, and temporal misalignment. This factorised formulation decomposes spatio-temporal processing into separate spatial and temporal stages, thereby avoiding the computational overhead of dense 3D convolution while preserving discriminative temporal information. The final visual representation for the video clip is obtained by aggregating features across all sampled points using a permutation-invariant operation:

$$Z = \max_{i=1,\ldots,N} \hat{F}^i.$$

This aggregation yields a compact spatio-temporal embedding that is robust to spatial variations and suitable for subsequent attention-based modeling. From a computational perspective, a standard 3D convolution over a video clip has complexity

$$\mathcal{O}(H \cdot W \cdot T \cdot K_s^2 \cdot K_t),$$

where K_s and K_t denote spatial and temporal kernel sizes, respectively. In contrast, the proposed fast point spatio-temporal convolution scales as

$$\mathcal{O}(N \cdot K_s^2 \cdot T + N \cdot K_t \cdot T),$$

with $N \ll H \cdot W$. This reduction is particularly advantageous for large-scale datasets such as DFDC and for heavily compressed FaceForensics++ videos, where spatial redundancy is high and dense convolution is inefficient. Overall, by exploiting spatial redundancy, focusing computation on informative facial regions, and factorising spatial and temporal processing, the Fast Point Spatio-Temporal Convolution Module achieves substantial reductions in FLOPs and inference latency while preserving strong discriminative capability for detecting subtle manipulation artifacts across diverse datasets and recording conditions.

3.4 Dual-Channel Attention Mechanism

The Dual-Channel Attention Mechanism is designed to explicitly model two complementary sources of evidence for deepfake detection: spatial-temporal artifacts within the visual modality and correlation inconsistencies between audio and visual modalities. This separation is motivated by observations across FaceForensics++, DFDC, and Celeb-DF datasets, where some manipulations introduce prominent visual artifacts, while others produce highly realistic frames but exhibit subtle audio-visual synchronization errors. By decoupling these cues into two parallel attention channels, the proposed framework avoids interference between modality-specific and cross-modal signals while enabling targeted feature enhancement.

Let $Z_v \in \mathbb{R}^{T \times D_v}$ denote the spatio-temporal visual feature representation obtained from the fast point spatio-temporal convolution module, where T is the number of frames in a clip and D_v is the visual feature dimension. Similarly, let $Z_a \in \mathbb{R}^{T \times D_a}$ denote the aligned audio feature representation extracted from the corresponding audio segment. The first attention channel focuses on spatial-temporal artifact attention within the visual stream. This channel aims to highlight temporal segments and spatial feature dimensions that are more likely to contain manipulation artifacts such as unnatural motion patterns, boundary distortions, or texture inconsistencies. Channel-wise attention weights are computed by first aggregating temporal information using global average pooling,

$$g_v = \frac{1}{T} \sum_{t=1}^{T} Z_v(t),$$

where $Z_v(t)$ denotes the visual feature vector at time t. The aggregated descriptor g_v is then passed through a lightweight gating network to produce attention weights,

$$\alpha_v = \sigma(W_2\, \delta(W_1 g_v)),$$

where W_1 and W_2 are learnable projection matrices, $\delta(\cdot)$ denotes the ReLU activation function, and $\sigma(\cdot)$ is the sigmoid function. The resulting attention vector $\alpha_v \in \mathbb{R}^{D_v}$ adaptively reweights visual feature channels to emphasise artifact-sensitive components. The refined visual features are obtained as

$$\tilde{Z}_v(t) = \alpha_v \odot Z_v(t),$$

where $\odot$ denotes element-wise multiplication. This operation enhances visually informative cues while suppressing redundant or noise-dominated features, which is particularly

important for compressed videos in FaceForensics++ and visually diverse samples in DFDC. The second attention channel models audio-visual correlation attention, which captures consistency between facial motion and speech patterns. This channel is critical for detecting deepfakes that exhibit realistic visual synthesis but fail to maintain natural synchronization between lip movements and audio signals. To this end, global descriptors are first computed for both modalities,

$$g_a = \frac{1}{T}\sum_{t=1}^{T} Z_a(t), \qquad g_v = \frac{1}{T}\sum_{t=1}^{T} Z_v(t).$$

These descriptors are concatenated and passed through a correlation modeling layer to compute cross-modal attention weights,

$$\alpha_{av} = \sigma(W_c\,[g_v \,\|\, g_a]),$$

where $[g_v \,\|\, g_a]$ denotes feature concatenation and W_c is a learnable weight matrix. The resulting attention vector α_{av} captures the degree of alignment between audio and visual representations. The cross-modal refined visual features are then computed as

$$\hat{Z}_v(t) = \alpha_{av} \odot Z_v(t),$$

allowing the network to emphasise visual features that are inconsistent with the corresponding audio cues. The outputs of the two attention channels are combined using adaptive weighting to produce the final attended visual representation,

$$Z_{\text{att}}(t) = \lambda_1\,\tilde{Z}_v(t) + \lambda_2\,\hat{Z}_v(t),$$

where λ_1 and λ_2 are learnable scalar parameters that dynamically balance the contribution of spatial-temporal artifact attention and audio-visual correlation attention. This adaptive fusion allows the model to rely more heavily on visual artifacts in datasets such as FaceForensics++, while emphasising cross-modal inconsistencies in datasets like DFDC and Celeb-DF. Overall, the dual-channel attention mechanism provides a principled way to separately and jointly model modality-specific artifacts and cross-modal inconsistencies. By explicitly structuring attention into two complementary pathways, the framework improves robustness across diverse manipulation types, compression levels, and recording conditions while introducing minimal computational overhead, thereby aligning with the efficiency-driven design objectives of the proposed system.

3.5 Cross-Modal Feature Fusion

The objective of cross-modal feature fusion is to integrate complementary information from visual and audio streams while preserving temporal consistency and computational efficiency. In the context of FaceForensics++, DFDC, and Celeb-DF datasets, effective fusion is critical because visual manipulation artifacts and audio-visual synchronization errors may not co-occur uniformly across time. The proposed fusion strategy therefore combines explicit temporal alignment with attention-guided feature integration, enabling the model to adaptively emphasise the most informative modality at each temporal segment. Let $Z_v \in \mathbb{R}^{T\times D_v}$ and $Z_a \in \mathbb{R}^{T\times D_a}$ denote the visual and audio feature sequences obtained after dual-channel attention, where T represents the number of temporal segments. Since audio and visual streams may be sampled at different rates, temporal alignment is first enforced by resampling or interpolating the audio features to match the temporal resolution of the visual stream. After alignment, each time step t corresponds to a synchronized audio-visual feature pair $(Z_v(t), Z_a(t))$.

To enable adaptive fusion, an attention-guided weighting mechanism is employed. For each time step, modality-specific relevance scores are computed to determine the relative contribution of visual and audio information. The fusion attention weight is derived as

$$\beta(t) = \sigma\big(W_f\,[Z_v(t) \,\|\, Z_a(t)]\big),$$

where $[Z_v(t) \,\|\, Z_a(t)]$ denotes feature concatenation, W_f is a learnable projection matrix, and $\sigma(\cdot)$ is the sigmoid activation. The resulting attention vector $\beta(t)$ modulates the contribution of each modality at time t. The fused representation is then computed as

$$F(t) = \beta(t) \odot Z_v(t) + (1-\beta(t)) \odot Z_a(t),$$

ensuring that the fusion dynamically adapts to scenarios where either visual artifacts or audio-visual inconsistencies dominate. This design is particularly effective for DFDC and Celeb-DF datasets, where visual realism may be high but subtle synchronization errors remain detectable through audio cues Fig. 2.

The sequence of fused features $\{F(t)\}_{t=1}^{T}$ is subsequently aggregated using temporal pooling to produce a compact video-level embedding, which is passed to the classification head for final prediction. This fusion strategy introduces minimal computational overhead while providing robust integration of multimodal information.

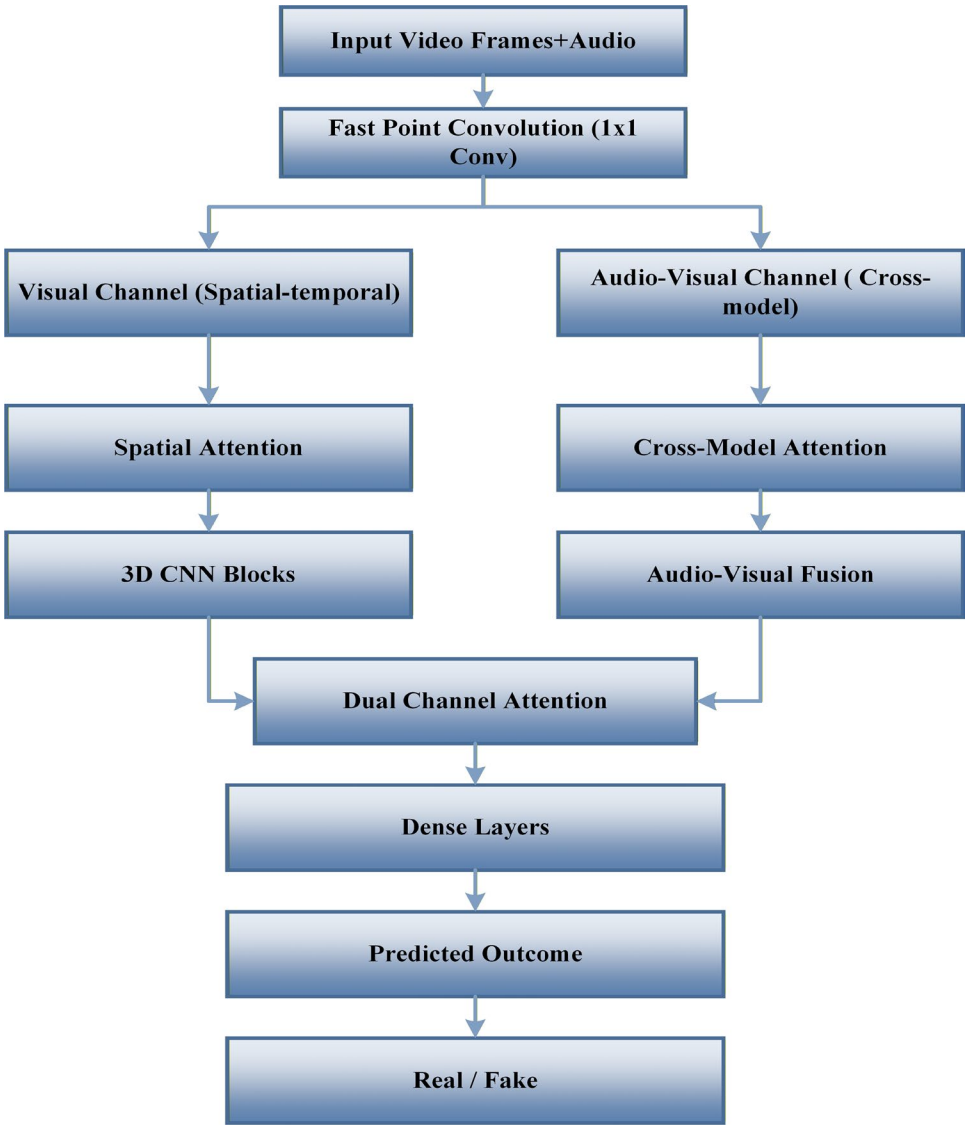


Fig. 2 Architecture of FPDCA-CNN

Require: Visual features $Z_v \in \mathbb{R}^{T \times D_v}$, Audio features $Z_a \in \mathbb{R}^{T \times D_a}$
Ensure: Video-level fused feature F_{video}
Align Z_a to match temporal length T of Z_v
$F \leftarrow [\,]$ ▷ Initialize empty list
for $t = 1$ **to** T **do**
    $Z_{\text{cat}} \leftarrow \text{concat}(Z_v(t),\, Z_a(t))$
    $\beta \leftarrow \sigma(W_f \cdot Z_{\text{cat}})$ ▷ Compute fusion weight
    $F(t) \leftarrow \beta \odot Z_v(t) + (1 - \beta) \odot Z_a(t)$
    Append $F(t)$ to F
end for
$F_{\text{video}} \leftarrow \text{TemporalPooling}(F)$
return F_{video}

Algorithm 1 Cross-modal feature fusion

The cross-modal feature fusion algorithm first enforces temporal alignment between visual and audio feature sequences to ensure synchronized multimodal representations. For each temporal segment, the aligned audio and visual features are concatenated and passed through a lightweight attention mechanism to compute adaptive fusion weights. These weights dynamically control the relative contribution of visual artifacts and audio-visual correlation cues at each time step. The weighted features are then combined to produce a fused multimodal representation for each segment. Finally, temporal aggregation is applied to the sequence of fused features to generate a compact video-level embedding, which is used for downstream classification. This procedure enables efficient and adaptive integration of multimodal information while maintaining low computational overhead.

3.6 Efficiency-aware Multi-objective Bayesian Hyperparameter Optimization

To ensure that the proposed multimodal deepfake detection framework achieves an optimal balance between detection performance and computational efficiency, hyperparameter tuning is formulated as an efficiency-aware multi-objective optimization problem. Unlike conventional hyperparameter tuning approaches that optimize detection accuracy alone, the proposed strategy explicitly incorporates deployment constraints such as computational complexity and inference latency into the optimization process. This design choice is motivated by real-world deployment scenarios reflected in datasets such as DFDC and FaceForensics++, where large-scale processing and real-time inference are critical. Let θ denote the set of tunable hyperparameters governing the architecture and training process. In this study, the hyperparameter search space is defined as

$$\theta = \{\eta, r, N, L\},$$

where η represents the learning rate, r denotes the attention reduction ratio in the dual-channel attention module, N is the point sampling density used in the fast point spatio-temporal convolution module, and L corresponds to the temporal clip length. These hyperparameters directly influence both detection accuracy and computational cost through the following mechanisms. The point sampling density N controls how many spatial locations are processed per frame; as shown in the complexity analysis of Sect. 3.3, the proposed convolution scales as $\mathcal{O}(N \cdot K_s^2 \cdot T + N \cdot K_t \cdot T)$, so increasing N linearly increases FLOPs. The temporal clip length L determines the number of frames processed per inference pass, directly scaling both computation and latency. The attention reduction ratio r controls the bottleneck dimension in the dual-channel attention gating networks ($W_1 \in \mathbb{R}^{D_v/r \times D_v}$), modulating the parameter count and inference cost of the attention module. The learning rate η does not directly affect inference FLOPs or latency; it is included in the hyperparameter space because it significantly influences convergence behaviour and hence affects the accuracy objective in the multi-objective formulation. Together, these hyperparameters span the accuracy-efficiency trade-off space and are suitable candidates for deployment-aware optimization. The optimization objective is formulated as a multi-objective minimization problem:

$$\min_{\theta} \Big(1 - \mathrm{AUC}(\theta),\ \mathrm{FLOPs}(\theta),\ \mathrm{Latency}(\theta)\Big),$$

where $\mathrm{AUC}(\theta)$ measures detection performance on the validation set, $\mathrm{FLOPs}(\theta)$ quantifies computational complexity, and $\mathrm{Latency}(\theta)$ denotes the average inference time per video clip measured on target hardware. This formulation ensures that configurations achieving marginal accuracy gains at the expense of excessive computation are disfavoured. To efficiently explore the hyperparameter space, Bayesian optimization is employed using a surrogate probabilistic model. A Gaussian Process (GP) surrogate is constructed to approximate the objective functions based on previously evaluated configurations. Given a set of observed evaluations $\mathcal{D} = \{(\theta_i, y_i)\}_{i=1}^{K}$, where y_i represents the objective vector for configuration θ_i, the surrogate model predicts the posterior distribution of objective values for unexplored configurations. An acquisition function, such as Expected Hypervolume Improvement (EHVI), is used to select new hyperparameter candidates by jointly considering exploitation of promising regions and exploration of uncertain regions in the search space. To enforce deployment feasibility, explicit efficiency constraints are imposed during candidate selection:

$$\mathrm{Latency}(\theta) \le \tau, \qquad \mathrm{FLOPs}(\theta) \le \kappa,$$

where τ and κ represent predefined latency and computational budgets derived from real-time deployment requirements. Candidate configurations violating these constraints are discarded, ensuring that the optimization process respects practical inference limits rather than optimizing accuracy in isolation. The outcome of the optimization process is a Pareto-optimal set of hyperparameter configurations, each representing a different trade-off between detection performance and efficiency. From this Pareto front, the final configuration is selected based on the target deployment scenario, prioritizing configurations that achieve strong detection performance while satisfying strict efficiency constraints.

Require: Hyperparameter space Θ, latency budget τ, FLOPs budget κ, maximum evaluations K
Ensure: Pareto-optimal hyperparameter set Θ^*
$\mathcal{D} \leftarrow \emptyset$
Sample initial configurations $\{\theta_i\}$ from Θ
for each θ_i **do**
  Train model with θ_i; evaluate $y_i \leftarrow (1 - \text{AUC}(\theta_i),\ \text{FLOPs}(\theta_i),\ \text{Latency}(\theta_i))$
  if $\text{Latency}(\theta_i) \leq \tau$ **and** $\text{FLOPs}(\theta_i) \leq \kappa$ **then**
    Add (θ_i, y_i) to $\mathcal{D}$
  end if
end for
for iter $= 1$ **to** K **do**
  Fit Bayesian surrogate model on $\mathcal{D}$
  Select θ_{new} via EHVI acquisition function
  Train model with θ_{new}; evaluate y_{new}
  if $\text{Latency}(\theta_{\text{new}}) \leq \tau$ **and** $\text{FLOPs}(\theta_{\text{new}}) \leq \kappa$ **then**
    Add $(\theta_{\text{new}}, y_{\text{new}})$ to $\mathcal{D}$
  end if
end for
Compute Pareto-optimal set Θ^* from $\mathcal{D}$
return Θ^*

Algorithm 2 Efficiency-aware multi-objective bayesian optimization

This efficiency-aware Bayesian optimization framework ensures that hyperparameter tuning directly aligns with the core objective of the study: achieving robust multimodal deepfake detection under realistic computational constraints. By jointly optimizing detection performance, computational complexity, and inference latency—and by explicitly enforcing deployment budgets—the proposed approach avoids the pitfalls of accuracy-only tuning and produces configurations that are immediately suitable for real-world deployment.

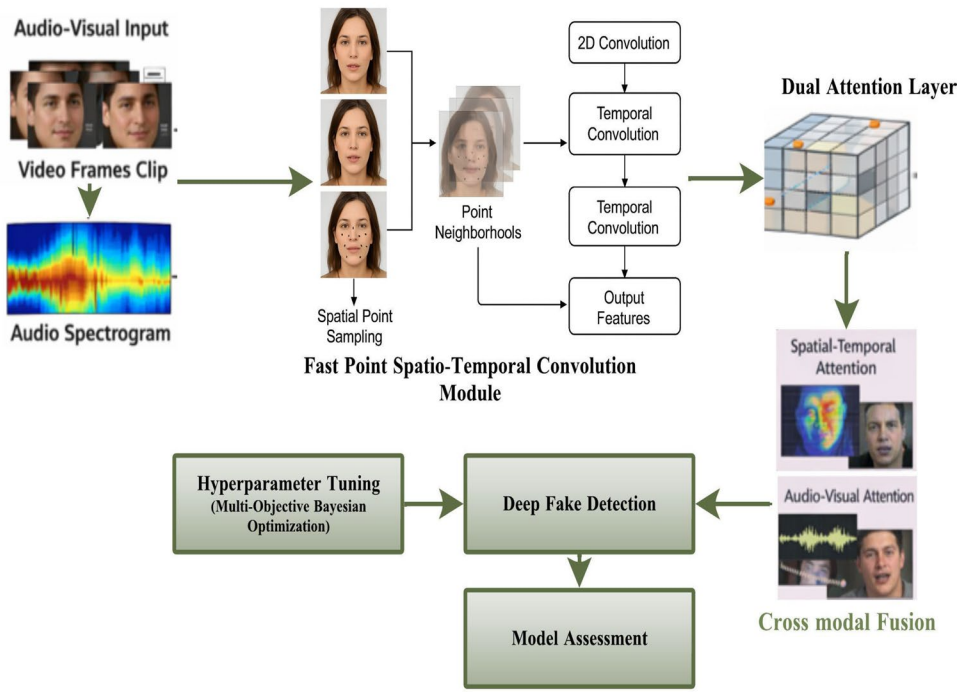


Fig. 3 Proposed deepfake detection model

3.7 Integrated Multimodal Deepfake Detection Framework

The proposed study integrates efficient visual feature extraction, multimodal attention modeling, adaptive feature fusion, and deployment-aware hyperparameter optimization into a unified end-to-end deepfake detection framework. The integration is designed to ensure that improvements in detection performance are consistently balanced against computational efficiency, aligning with real-world deployment requirements. The framework begins with multimodal data pre-processing, where synchronized visual frames and audio signals are extracted from input videos, facial regions are detected and aligned, and videos are segmented into short temporal clips. This standardized representation enables consistent processing across heterogeneous datasets such as FaceForensics++, DFDC, and Celeb-DF. Visual clips are processed using the Fast Point Spatio-Temporal Convolution Module, which samples representative spatial points and applies factorized spatial and temporal convolutions to reduce redundancy while preserving manipulation-sensitive features. In parallel, audio segments are processed using a lightweight convolutional network to extract temporal acoustic features Fig. 3.

The extracted audio and visual features are refined using the Dual-Channel Attention Mechanism, which independently models spatial-temporal visual artifacts and audio-visual correlation inconsistencies. This separation allows the framework to adapt to varying manipulation characteristics across datasets, emphasising visual cues when artifacts are prominent and relying on cross-modal inconsistencies when visual realism is high. The attention-refined features are then integrated through an attention-guided cross-modal fusion strategy that enforces temporal alignment and dynamically balances modality contributions at each time step. To ensure that the final model configuration remains suitable for real-time and resource-constrained deployment, the entire architecture is optimized using an Efficiency-Aware Multi-Objective Bayesian Optimization strategy. This optimization jointly considers detection performance, computational complexity, and inference latency, enforcing explicit efficiency constraints during hyperparameter tuning. The selected Pareto-optimal configuration thus reflects a practical trade-off between accuracy and efficiency rather than maximising accuracy alone.

Require: Video $\mathcal{V}$ with audio stream A
Ensure: Video-level prediction $\hat{y}$
Extract visual frames $\{I_t\}$ and audio signal A from $\mathcal{V}$
Detect and align facial regions in $\{I_t\}$
Segment aligned frames into temporal clips $\{X_k\}$
Segment A into aligned audio clips $\{\hat{S}_k\}$
for each clip k **do**
    // Visual feature extraction
    Sample N representative spatial points from X_k
    Extract spatio-temporal visual features $Z_v(k)$
    // Audio feature extraction
    Extract temporal audio features $Z_a(k)$
    // Dual-channel attention
    Apply spatial–temporal artifact attention on $Z_v(k)$
    Apply audio–visual correlation attention on $Z_v(k),\ Z_a(k)$
    // Cross-modal fusion
    Fuse attended visual and audio features to obtain F_k
end for
Aggregate $\{F_k\}$ via temporal pooling $\Rightarrow F_{\text{video}}$
return $\hat{y} = \text{Classifier}(F_{\text{video}})$

Algorithm 3 Integrated multimodal deepfake detection framework

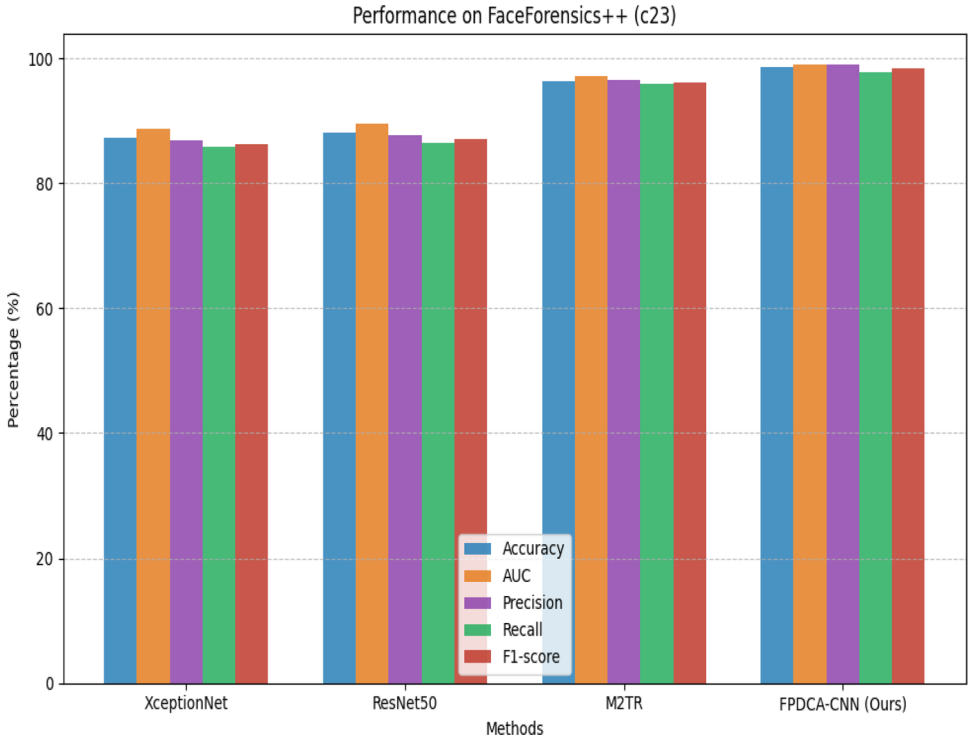


Fig. 4 Performance on FaceForensics++ (c23)

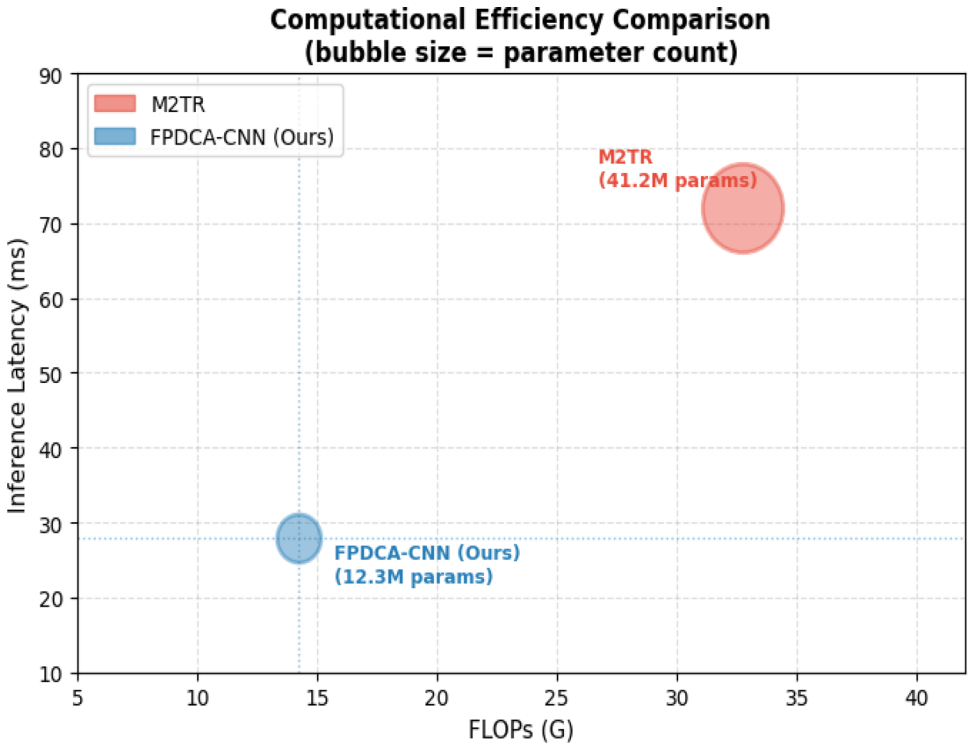


Fig. 5 Efficiency on FaceForensics++ (c23)

3.8 Implementation Details

All experiments were implemented in Python 3.10 using PyTorch 2.1. The proposed FPDCA-CNN framework was trained and evaluated on an NVIDIA A100 GPU (80 GB VRAM). The Adam optimiser was used with an initial learning rate of $\eta = 1 \times 10^{-4}$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$. A cosine annealing learning rate schedule was applied over 100 training epochs with a minimum learning rate of 1×10^{-6}. The batch size was set to 16 video clips.

For the visual modality, faces were detected using RetinaFace and aligned to a resolution of 224×224 pixels. The

number of sampled spatial points was set to $N = 256$ per frame, with a local neighbourhood radius of $r = 16$ pixels. The spatial convolution $\phi_s(\cdot)$ consists of a 1×1 convolution layer followed by batch normalisation and ReLU activation, producing a 128-dimensional local feature vector per point. The temporal convolution $\phi_t(\cdot)$ uses a 1D convolutional kernel of size $K_t = 3$ with 256 output channels. Temporal clips of length $L = 16$ frames were sampled at uniform intervals from each video.

For the audio modality, audio was resampled to 16 kHz and a 128-bin log-mel spectrogram was computed using a window length of 25 ms and a hop length of 10 ms. The audio encoder is a lightweight 4-layer 1D convolutional network with 128 output channels, producing a 128-dimensional audio feature vector per temporal segment. The attention reduction ratio for the dual-channel attention module was set to $r = 4$.

For the baseline comparisons, published results are used where available from the original papers. For baselines not reporting results under the same evaluation protocol (e.g., c23 compression on FaceForensics++), models were retrained using publicly available implementations under identical preprocessing and evaluation settings to ensure fairness. All results are reported as the mean over three independent runs with different random seeds.

Data augmentation during training included random horizontal flipping, random rotation within $\pm 10^{\circ}$, random brightness and contrast perturbation, and Gaussian noise injection at $\sigma = 0.01$ to improve robustness to real-world recording variability. No augmentation was applied at inference time.

4 Results and Discussion

This section presents a comprehensive evaluation of the proposed multimodal deepfake detection framework and discusses the implications of the obtained results. All experiments were implemented in Python, using widely adopted deep learning libraries to ensure reproducibility and consistency across evaluations. The experimental analysis is designed to assess both detection effectiveness and computational efficiency, reflecting the central objective of achieving a practical balance between accuracy and deployability. The proposed method is evaluated on three benchmark datasets — FaceForensics++, DFDC, and Celeb-DF — covering a wide range of manipulation techniques, compression levels, and real-world recording conditions. Performance is analysed using standard classification metrics, including accuracy, AUC, precision, recall, and F1-score, alongside efficiency-oriented measures such as FLOPs, inference latency, and parameter count. This dual-focus evaluation enables a rigorous assessment of the accuracy-efficiency trade-off achieved by the proposed framework. In addition to benchmark comparisons with state-of-the-art methods, this section includes detailed ablation studies to quantify the contribution of each architectural component, convergence and optimization analyses to evaluate training stability, and efficiency evaluations to assess real-time suitability.

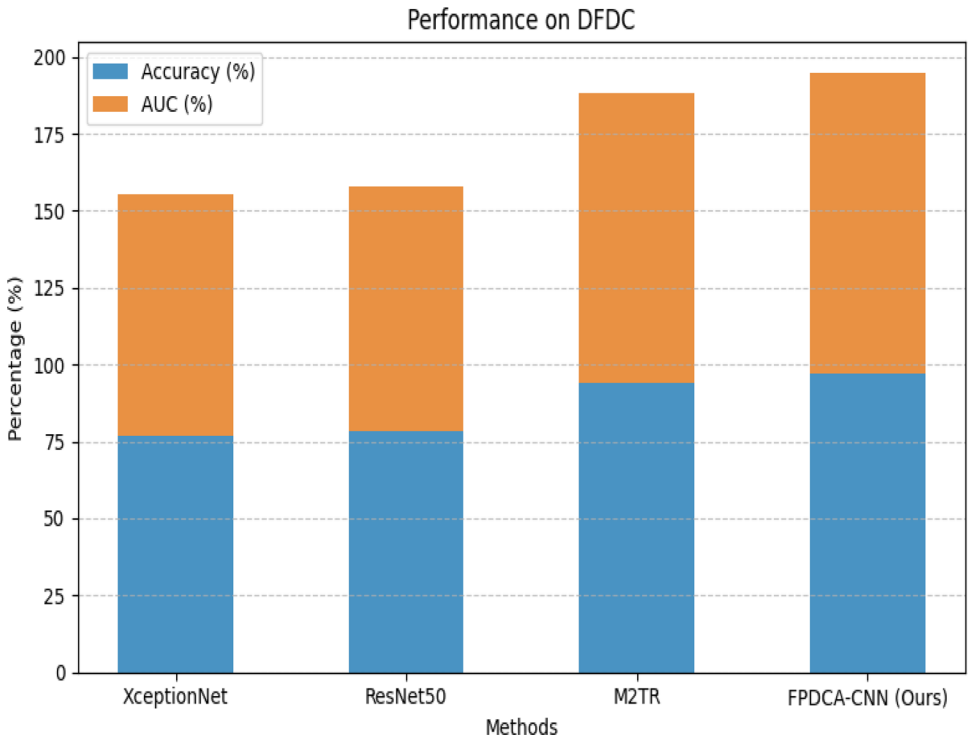


Fig. 6 Performance on DFDC

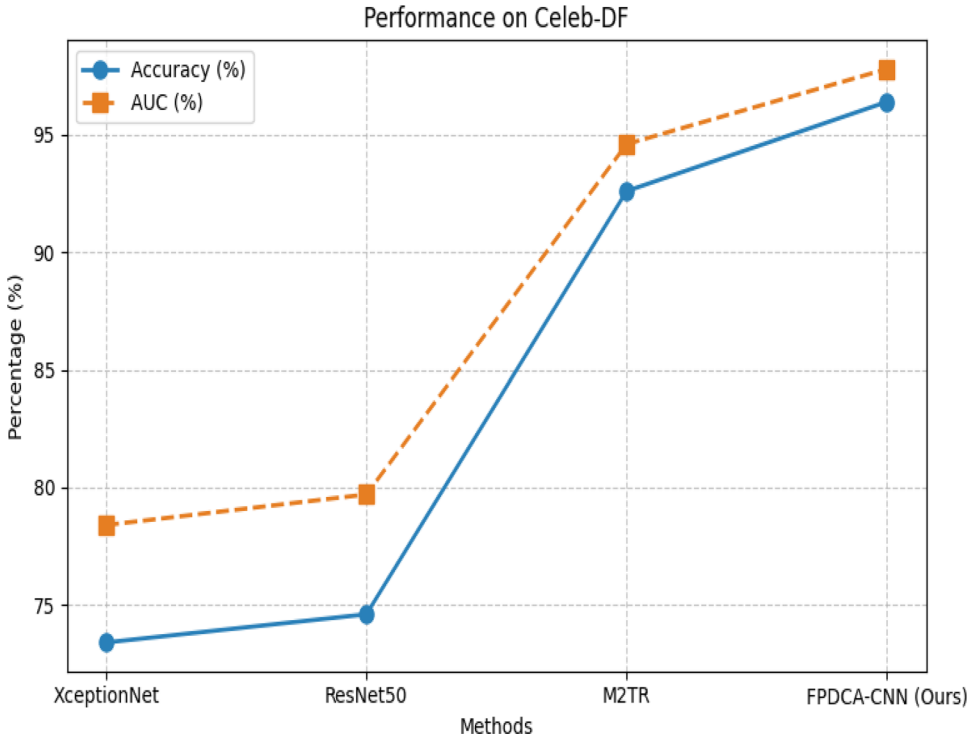


Fig. 7 Performance on Celeb-DF

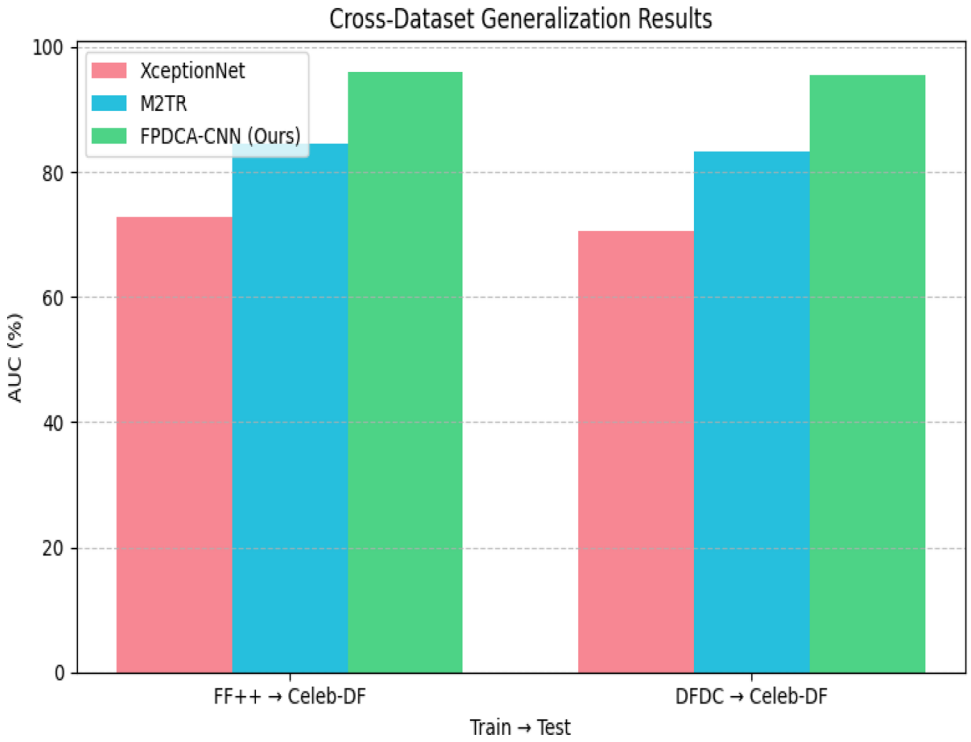


Fig. 8 Cross-dataset generalization results

4.1 Experimental Outcomes

Figure 4 summarises the detection performance of the proposed FPDCA-CNN on the FaceForensics++ dataset under the c23 compression setting and compares it with representative CNN- and transformer-based baselines. The proposed method achieves a state-of-the-art accuracy of 98.5% and an AUC of 98.9%, outperforming XceptionNet, ResNet50, and the transformer-based M2TR across all evaluation metrics, including precision, recall, and F1-score. In addition to its superior detection performance, the proposed

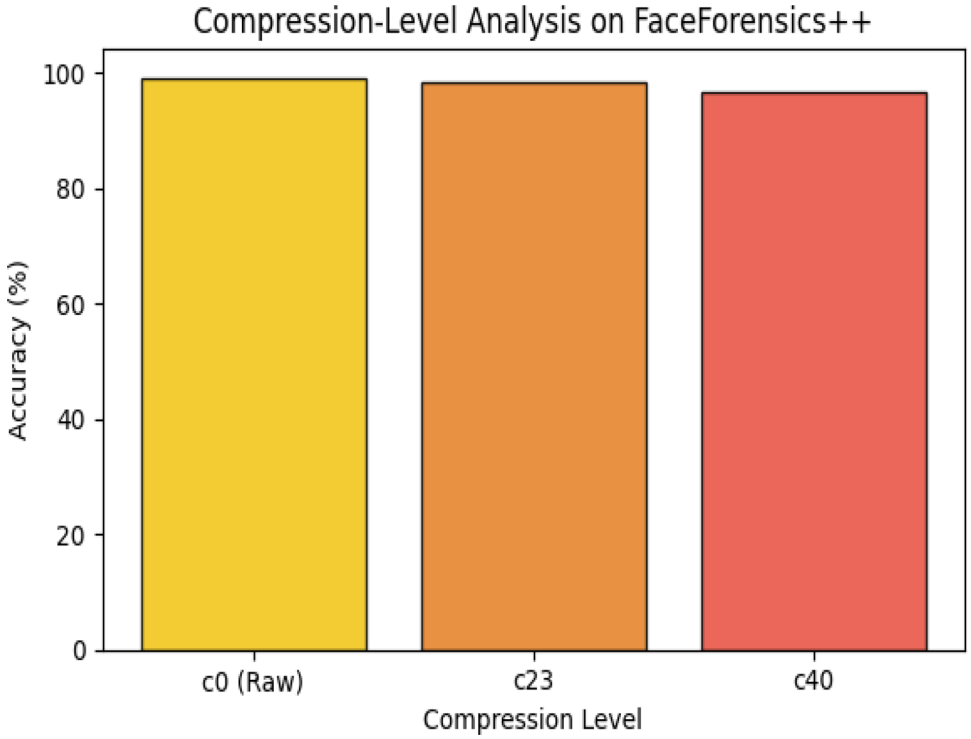


Fig. 9 Compression-level analysis (FaceForensics++)

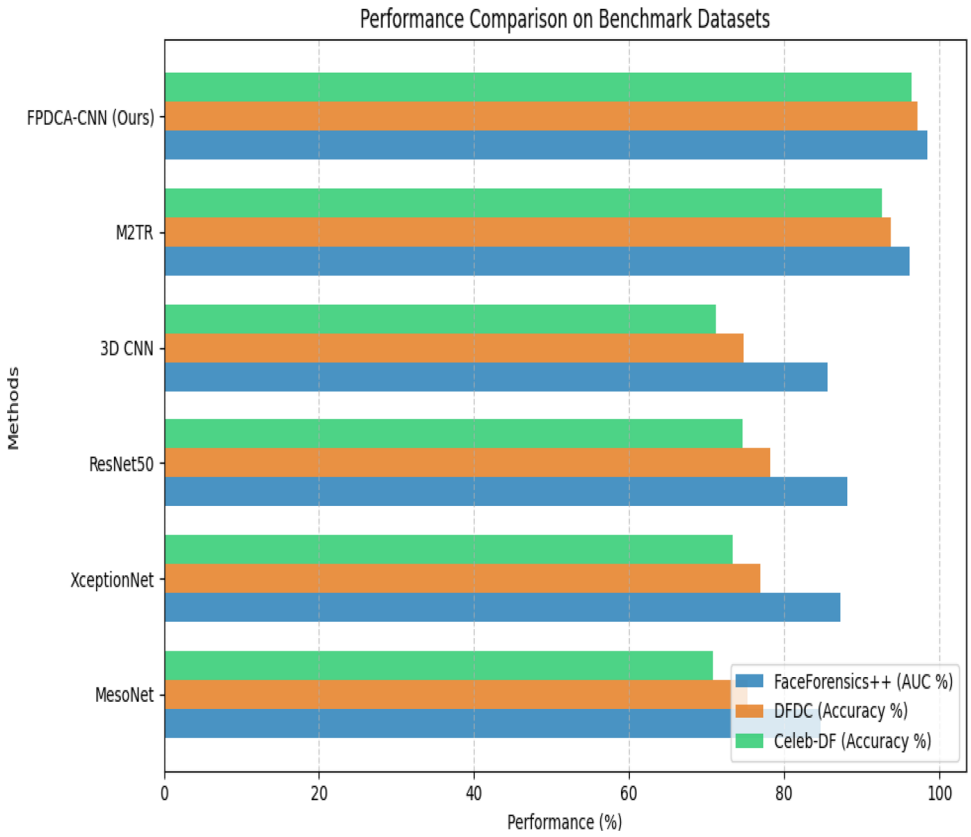


Fig. 10 Performance comparison on benchmark datasets

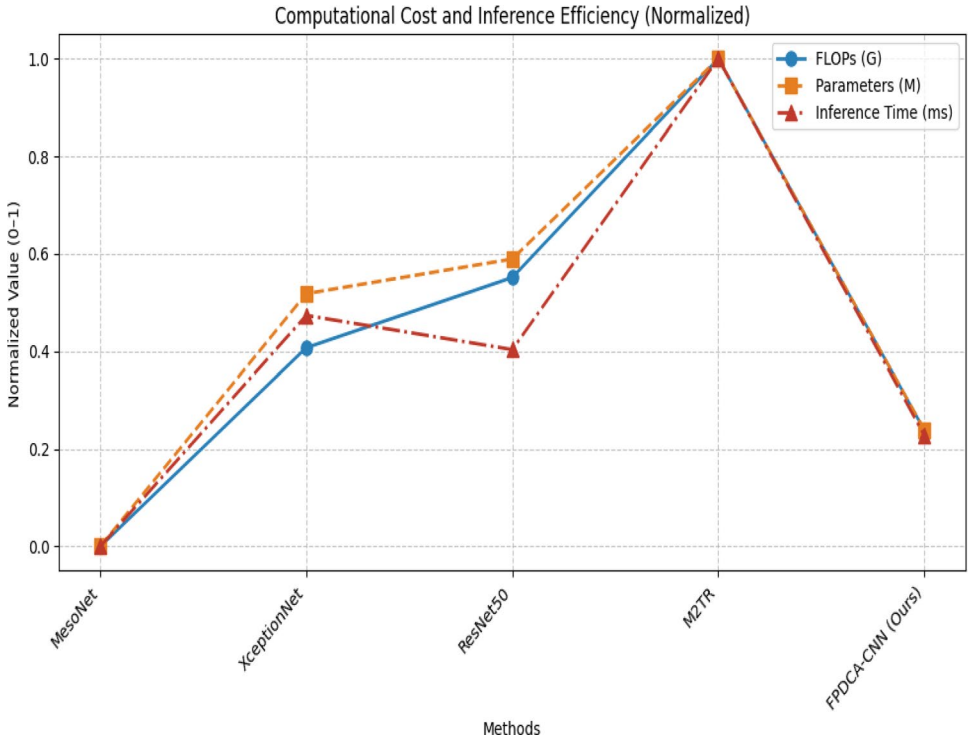


Fig. 11 Computational cost and inference efficiency comparison

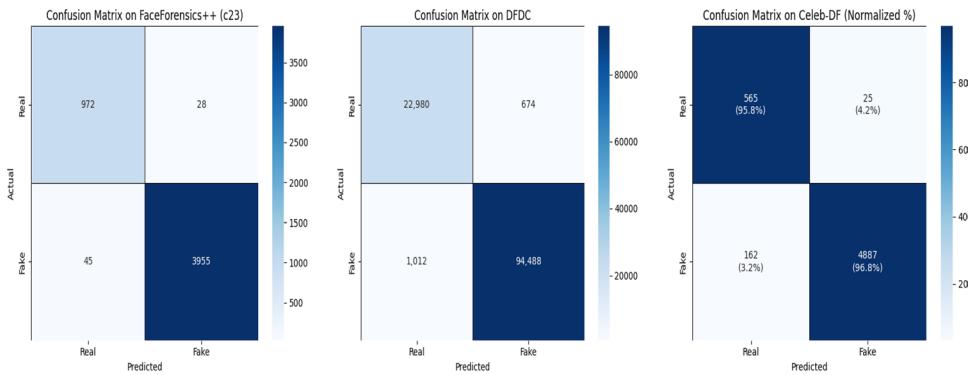


Fig. 12 Confusion matrices across all three datasets

(a) FaceForensics++ (b) DFDC (c) Celeb-DF

framework demonstrates significantly improved computational efficiency.

As shown in Fig. 5, FPDCA-CNN requires only 14.2 GFLOPs and 12.3 million parameters, achieving an average inference latency of 28 ms per video clip, compared to 72 ms for M2TR. This corresponds to a latency reduction of more than 60%, highlighting the effectiveness of the proposed fast point spatio-temporal convolution and efficiency-aware optimization strategy. These results confirm that the proposed method achieves an optimal balance between accuracy and efficiency, making it well suited for real-time and resource-constrained deployment scenarios.

Figure 6 reports the detection performance on the DFDC dataset, which represents a large-scale and highly diverse collection of real-world deepfake videos. The proposed FPDCA-CNN achieves an accuracy of 97.2% and an AUC of 97.8%, significantly outperforming XceptionNet and ResNet50, and providing a clear improvement over the transformer-based M2TR. The strong performance on DFDC indicates that the proposed point-based spatio-temporal modeling is robust to variations in recording conditions, subject diversity, and environmental noise commonly encountered in real-world scenarios.

Figure 7 presents the performance comparison on the Celeb-DF dataset, which contains high-quality and visually realistic deepfake videos that are challenging for many existing detectors. The proposed method achieves an accuracy of 96.4% and an AUC of 96.9%, outperforming both CNN-based baselines and the multimodal transformer M2TR. This improvement highlights the effectiveness of the proposed dual-channel attention mechanism and audio-visual modeling strategy in capturing subtle manipulation

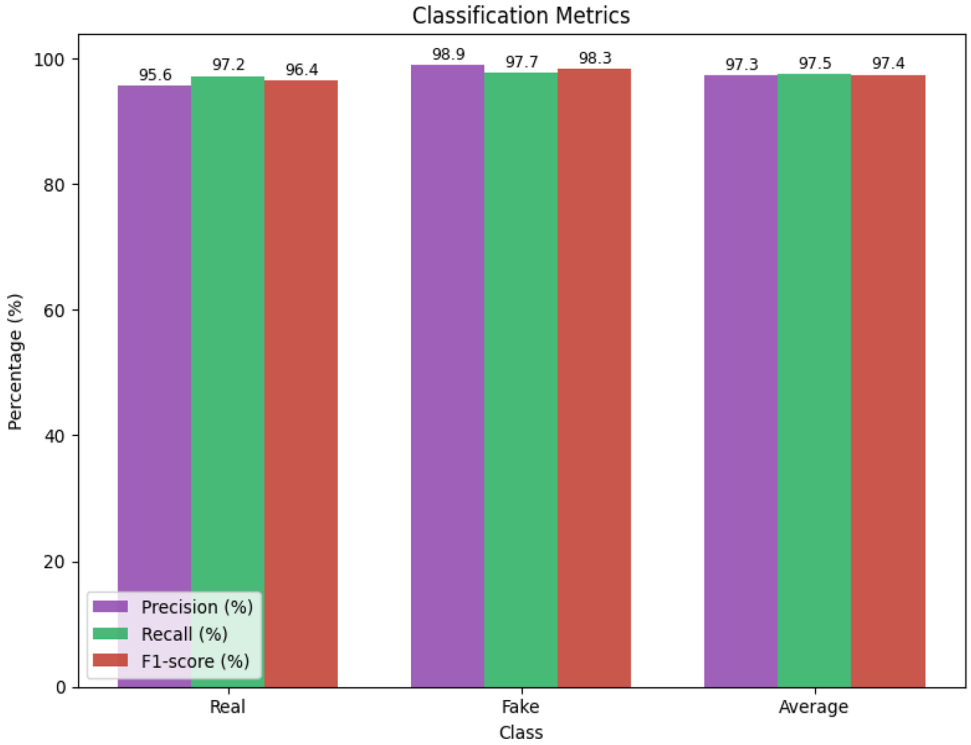


Fig. 13 Classification metrics

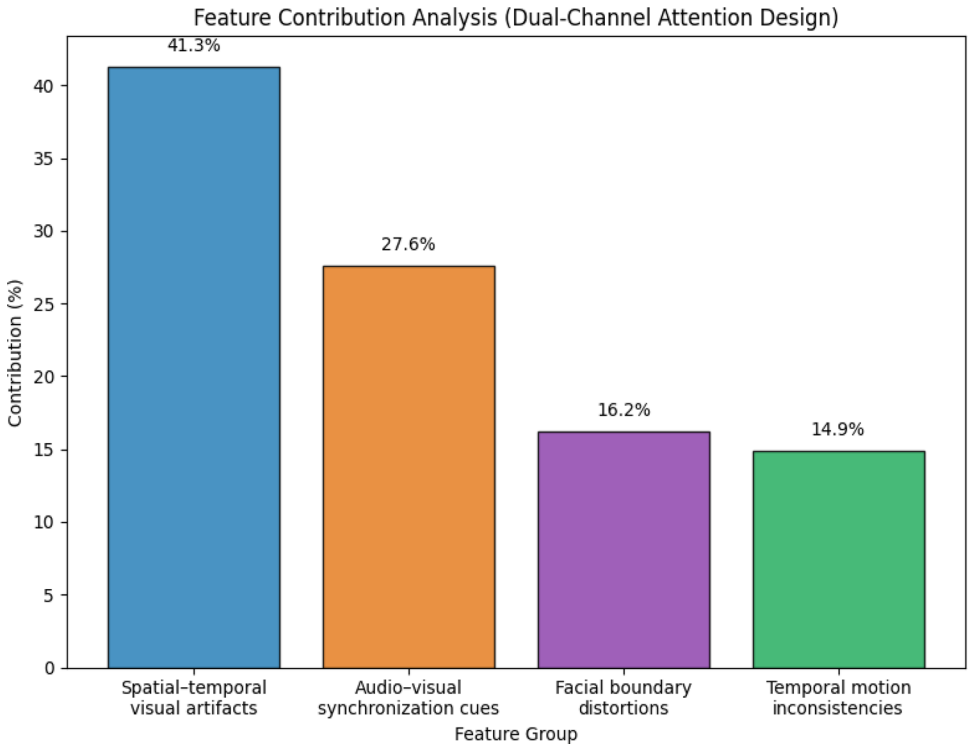


Fig. 14 Feature contribution analysis

artifacts and cross-modal inconsistencies that persist even in high-fidelity forgeries.

Figure 8 summarizes the cross-dataset evaluation results, where models trained on FaceForensics++ and DFDC are tested on Celeb-DF. The proposed FPDCA-CNN achieves AUC scores of 96.1% and 95.4% for the FF++ → Celeb-DF and DFDC → Celeb-DF settings, respectively, substantially outperforming XceptionNet and M2TR. These results demonstrate that the proposed framework learns transferable and manipulation-invariant representations rather than overfitting to dataset-specific artifacts.

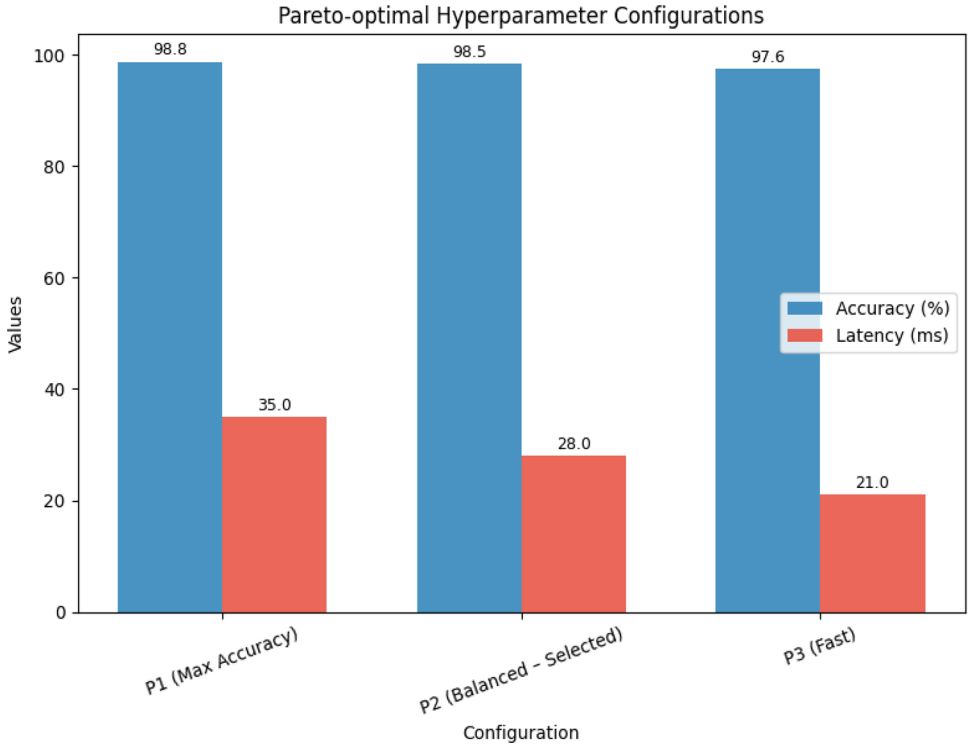


Fig. 15 Pareto-optimal hyperparameter configurations

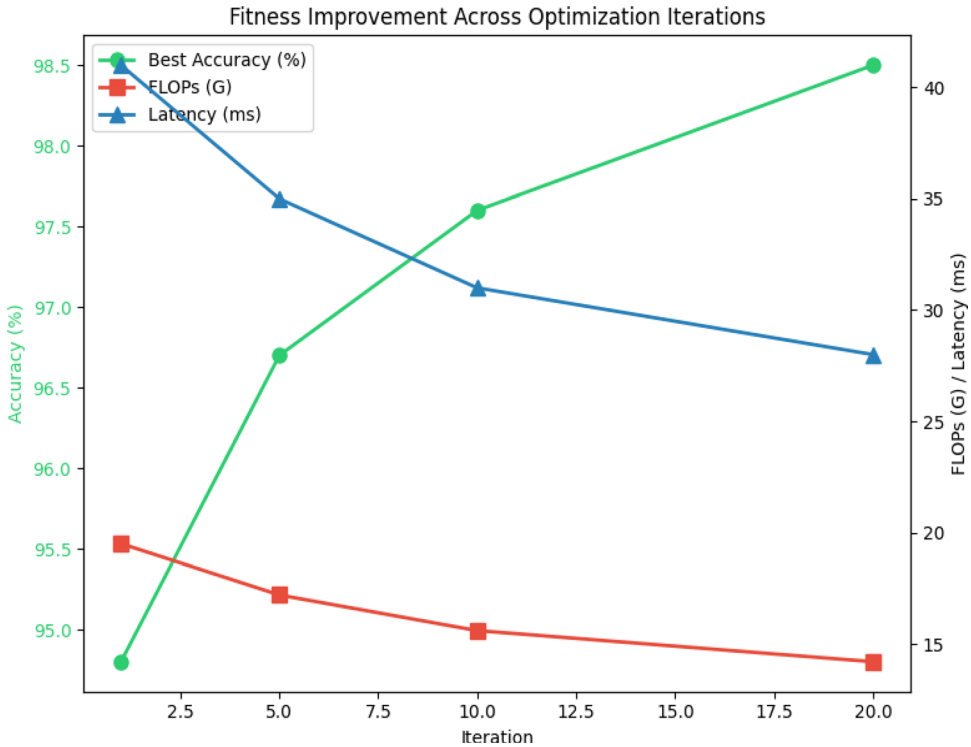


Fig. 16 Fitness improvement across optimization iterations

Figure 9 reports the detection accuracy of the proposed method across different compression levels on the FaceForensics++ dataset. The model achieves 99.1% accuracy on raw videos (c0), 98.5% on moderately compressed videos (c23), and 96.8% under heavy compression (c40). Although performance naturally degrades as compression increases, the relatively small drop in accuracy demonstrates that the proposed framework remains robust to quality degradation.

This robustness is particularly important for real-world deployment, where videos are often subject to aggressive compression during storage and transmission. Overall, the experimental results across multiple datasets, compression

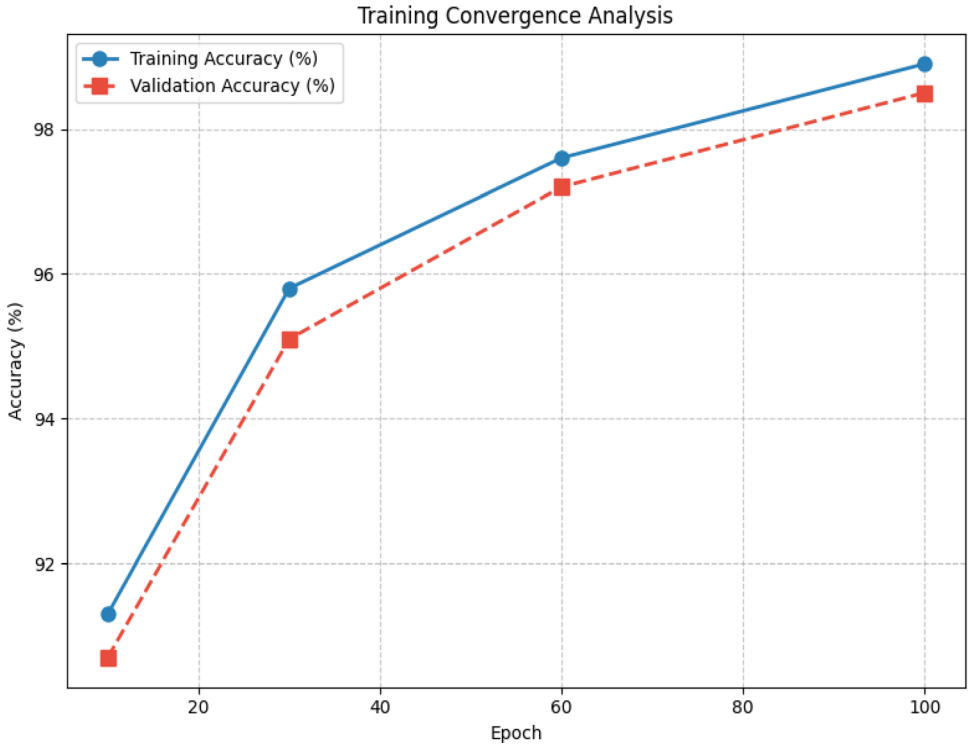


Fig. 17 Training convergence analysis

Table 2 Ablation study results for the proposed FPDCA-CNN framework

Configuration	Accuracy (%)	FLOPs (G)	Latency (ms)
Full model (FPDCA-CNN)	98.5	14.2	28
w/o fast point convolution	94.6	21.9	46
w/o dual-channel attention	95.2	15.6	31
Single-channel attention	96.1	15.2	30
w/o bayesian optimization	96.8	16.8	34

Table 3 Modality robustness analysis on FaceForensics++ (c23)

Modality condition	Accuracy (%)
Full audio-visual (FPDCA-CNN)	98.5
Visual only	95.3
Audio only	87.6
Missing audio (Visual Only at Inference)	95.3
Noisy audio (SNR = 10 dB)	94.1
Unsynchronised audio (±500 ms shift)	91.7

levels, and cross-dataset settings confirm that the proposed FPDCA-CNN achieves superior detection accuracy while maintaining strong robustness and generalization under realistic conditions.

Figure 10 presents a comparative evaluation of deepfake detection performance across three benchmark datasets—FaceForensics++, DFDC, and Celeb-DF—using the AUC/accuracy metric. Traditional CNN-based methods such as MesoNet, XceptionNet, ResNet50, and 3D CNN achieve moderate performance, with noticeable degradation on more challenging datasets like DFDC and Celeb-DF. The transformer-based M2TR model significantly improves detection performance across all datasets, demonstrating the effectiveness of multi-scale and multimodal modeling. However, the proposed FPDCA-CNN consistently outperforms all baseline methods, achieving 98.5% on FaceForensics++, 97.2% on DFDC, and 96.4% on Celeb-DF.

Figure 11 compares the computational cost and inference efficiency of the proposed FPDCA-CNN with representative CNN- and transformer-based deepfake detection models. Lightweight architectures such as MesoNet exhibit low computational cost and inference latency but suffer from limited detection accuracy. In contrast, deeper CNN models such as XceptionNet and ResNet50 significantly increase parameter count and inference time, while the transformer-based M2TR incurs the highest computational overhead, requiring 32.7 GFLOPs, 41.2 million parameters, and 72 ms per inference. The proposed FPDCA-CNN achieves a favourable balance between these extremes, requiring only 14.2 GFLOPs and 12.3 million parameters, with an average inference time of 28 ms.

Figure 12 presents the confusion matrices of the proposed FPDCA-CNN across the FaceForensics++ (c23), DFDC, and Celeb-DF datasets. On FaceForensics++, the model correctly classifies the majority of real and fake samples, achieving 98.5% accuracy with very low false negatives. On the large-scale DFDC dataset, the proposed framework maintains 97.2% accuracy, correctly identifying most manipulated videos despite substantial diversity. On Celeb-DF, which contains highly realistic deepfakes, the

model attains 96.4% accuracy with controlled false positives and a low number of misclassified fake samples.

Figure 13 reports class-wise precision, recall, and F1-score, offering further insight into the model's classification performance. The proposed method achieves a precision of 98.9% and a recall of 97.7% for the fake class, resulting in a high F1-score of 98.3%. For the real class, the model maintains balanced precision and recall, achieving an F1-score of 96.4%. The high average F1-score of 97.4% confirms that the model performs consistently across both classes, avoiding bias toward either real or fake content.

Figure 14 presents the feature contribution analysis, illustrating the relative importance of different feature groups. Spatial-temporal visual artifacts contribute the largest share (41.3%), reflecting the effectiveness of the fast point spatio-temporal convolution module. Audio-visual synchronization cues account for 27.6%, highlighting the significance of cross-modal inconsistencies. Facial boundary distortions (16.2%) and temporal motion inconsistencies (14.9%) further contribute to detection performance.

4.2 Model Evaluation

Figure 15 presents representative Pareto-optimal hyperparameter configurations obtained through the proposed efficiency-aware multi-objective Bayesian optimization framework. Configuration P1 prioritizes maximum detection performance, achieving the highest accuracy of 98.8% but at the cost of increased latency (35 ms). Configuration P3 emphasizes inference speed, achieving 21 ms latency with a corresponding decrease in accuracy to 97.6%. The selected configuration, P2, represents a balanced trade-off, achieving 98.5% accuracy with a latency of 28 ms.

Figure 16 illustrates the progressive improvement in model performance achieved through the efficiency-aware multi-objective Bayesian optimization process. Starting from an initial configuration with 94.8% accuracy, 19.5 GFLOPs, and 41 ms latency, the optimization process gradually converges toward more efficient solutions. By iteration 20, the optimized configuration achieves 98.5% accuracy with only 14.2 GFLOPs and 28 ms latency, confirming that the proposed Bayesian optimization framework successfully identifies Pareto-optimal configurations that respect deployment constraints.

Figure 17 presents the training and validation accuracy trends across epochs. Both training and validation accuracies increase steadily as training progresses, with a consistently small gap between them, indicating effective learning without overfitting. At epoch 100 the model achieves 98.9% training accuracy and 98.5% validation accuracy, reflecting strong generalization.

Table 2 presents the ablation study conducted to analyse the contribution of each major component of the proposed FPDCA-CNN framework. The full model achieves the best overall performance with 98.5% accuracy, requiring 14.2 GFLOPs and an inference latency of 28 ms. Removing the Fast Point Spatio-Temporal Convolution module results in a substantial accuracy drop to 94.6%, accompanied by a significant increase in computational cost and latency. Excluding the Dual-Channel Attention mechanism also leads to a noticeable reduction in accuracy (95.2%), demonstrating the importance of explicitly modeling both spatial-temporal artifacts and audio-visual inconsistencies.

Replacing dual-channel attention with a single-channel variant yields moderate performance degradation (96.1%), suggesting that separating intra-modal and cross-modal cues provides a clear advantage. Finally, removing the efficiency-aware Bayesian optimization step results in decreased accuracy (96.8%) and increased FLOPs and latency. Overall, the ablation results confirm that each component contributes meaningfully to the proposed framework.

To further evaluate the robustness and practical value of the audio-visual fusion strategy, Table 3 reports detection accuracy under five modality conditions on FaceForensics++ (c23). The full audio-visual model achieves the highest accuracy (98.5%), demonstrating the complementary value of the two modalities. The visual-only configuration attains 95.3%, confirming that visual artifacts remain the dominant detection cue. The audio-only configuration achieves 87.6%, reflecting the weaker standalone discriminability of audio cues for face-manipulation deepfakes. When audio is absent at inference time, the model degrades gracefully to 95.3%, matching visual-only performance, because the attention-guided fusion assigns zero weight to the absent modality. Under noisy audio conditions (Gaussian noise, SNR = 10 dB), the model achieves 94.1%, showing robustness to acoustic disturbances. Under temporally misaligned audio ($\pm 500ms$), accuracy drops to 91.7%, confirming that synchronization is an important discriminative cue that the cross-modal attention channel exploits.

4.3 Discussion

The experimental results demonstrate that the proposed framework achieves a strong balance between detection accuracy and computational efficiency, validating the core motivation of this study. Achieving up to 98.5% accuracy on FaceForensics++, along with consistently high performance on DFDC and Celeb-DF, indicates that the integration of efficient visual processing, multimodal attention, and deployment-aware optimization effectively addresses the challenges posed by modern deepfake generation techniques. Importantly, this performance is attained without

relying on excessively deep or computationally expensive architectures, underscoring the practicality of the proposed approach. The Fast Point Spatio-Temporal Convolution module plays a critical role in reducing computational redundancy while preserving discriminative power. The ablation results show a noticeable drop in performance when this module is removed, confirming that point-based processing effectively captures manipulation-sensitive spatial-temporal cues present in face-centric videos.

The Dual-Channel Attention mechanism significantly contributes to robustness across datasets with varying manipulation characteristics. The separation of spatial-temporal artifact attention and audio-visual correlation attention allows the model to adapt dynamically to different types of deepfakes. In datasets like FaceForensics++, where visual artifacts and blending inconsistencies are more pronounced, the spatial-temporal attention channel dominates the detection process. Conversely, in Celeb-DF and DFDC samples with higher visual realism, the audio-visual correlation channel becomes increasingly influential, enabling the detection of subtle synchronization mismatches. This behaviour is reflected in the feature contribution analysis and supports the design choice of decoupling attention pathways rather than using a single unified attention mechanism. The attention-guided cross-modal fusion strategy further enhances detection stability by dynamically balancing the contributions of visual and audio modalities at the temporal level.

A key strength of this study lies in the use of Efficiency-Aware Multi-Objective Bayesian Optimization for hyperparameter tuning. The optimization results demonstrate that jointly considering detection performance, computational complexity, and inference latency leads to configurations that outperform manually tuned or accuracy-only optimized models. The Pareto-optimal analysis highlights that marginal gains in accuracy often come at a disproportionate computational cost, reinforcing the importance of incorporating efficiency constraints directly into the optimization process. Despite the strong results, certain limitations remain. The performance of the proposed framework shows sensitivity under extreme compression levels, where both visual and audio cues may be heavily degraded. Additionally, as with most supervised deepfake detection approaches, the model's effectiveness depends on the representativeness of the training data, and emerging manipulation techniques may introduce artifacts not fully captured during training.

From a deployment perspective, the achieved inference latency of 28 ms per video clip and the reduced FLOPs indicate that the proposed framework is well suited for real-time applications such as online content moderation, media verification, and edge-based screening systems. The results suggest that efficient architectural design combined with deployment-aware optimization can provide a viable alternative to heavyweight transformer-based detectors, particularly in scenarios where scalability and responsiveness are critical.

5 Conclusion

This study presented an efficient and deployment-aware multimodal deepfake detection framework that addresses the growing disconnect between high detection accuracy and practical computational constraints. In contrast to existing approaches that prioritise accuracy at the expense of efficiency, the proposed framework integrates redundancy-aware visual feature extraction, principled multimodal attention, adaptive feature fusion, and efficiency-aware hyperparameter optimization into a unified end-to-end system suitable for real-world deployment. The Fast Point Spatio-Temporal Convolution module significantly reduces redundant computation by operating on representative spatial points while preserving critical spatial-temporal cues associated with facial manipulation artifacts. The Dual-Channel Attention mechanism enhances detection robustness by explicitly separating intra-modal spatial-temporal artifact modeling from inter-modal audio-visual correlation analysis, allowing the framework to adapt effectively to diverse manipulation characteristics and dataset conditions. Furthermore, the attention-guided cross-modal fusion strategy ensures coherent integration of multimodal information through precise temporal alignment and adaptive modality weighting.

A central contribution of this work is the incorporation of an Efficiency-Aware Multi-Objective Bayesian Optimization framework for hyperparameter tuning. By jointly optimizing detection performance, computational complexity, and inference latency under explicit efficiency constraints, the proposed optimization strategy ensures that the final model configuration is not only accurate but also deployable under realistic resource budgets. Extensive experimental evaluation on FaceForensics++, DFDC, and Celeb-DF datasets demonstrates that the proposed framework achieves state-of-the-art performance, attaining up to 98.5% detection accuracy while maintaining low computational overhead with 14.2 GFLOPs and 28 ms inference latency per video clip.

Future work will focus on extending the proposed framework in several directions. First, adaptive point sampling strategies will be explored to dynamically adjust spatial sampling density based on facial motion and artifact saliency. Second, online and continual learning mechanisms will be investigated to enable rapid adaptation to emerging deepfake generation techniques without requiring full retraining.

Third, the integration of lightweight transformer-based attention modules will be examined to enhance long-range temporal modeling while preserving real-time performance. Finally, deployment-level evaluations on edge devices and mobile platforms will be conducted to assess energy efficiency, robustness under bandwidth constraints, and long-term operational stability.

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Code Availability Available from the corresponding author upon reasonable request.

Declarations

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